

Database Management Systems

CPS 3740

Lecture 17 (Optional)

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Outline

- Continue last lecture
- Receive multiple records in PHP from HTML CGI
 - Insert, delete and update operations
- Image & project
- Online order discussion (extra)
- Last withdraw date soon

Store Images in MySQL

- Blob – Binary Large Object
- Store image in binary large object (**blob**) data type

```
CREATE TABLE images(  
  id int(11) NOT NULL AUTO_INCREMENT,  
  image blob NOT NULL,  
  created date,  
  PRIMARY KEY (id)) ;
```

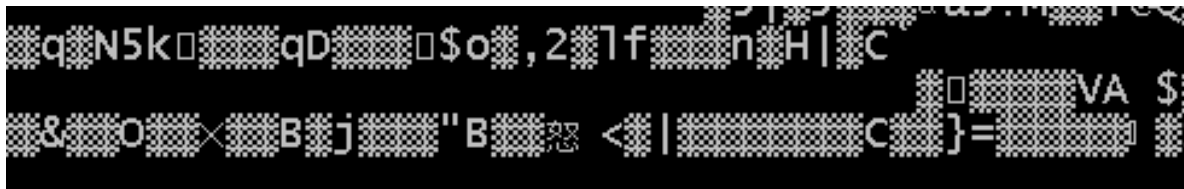
- Insert an image into the table.

```
insert into images(image, created) values  
(LOAD_FILE('/home/mysql/all_images/camera.jpg'),curdate());
```

Access Images in MySQL – terminal

- An binary image consists of pixels that requires graphical tools to display the content (pixels).
- Since Windows putty or Mac terminal is a text tool, it cannot display images correctly.
- The following SQL won't display the image on the command-line. It will show some random symbols.

**SELECT id, img, DOB FROM CPS3740.Customers
where id=1;**



Access Images in MySQL – Workbench #1

- Use workbench

`SELECT id, img, DOB FROM CPS3740.Customers where id=1;`

	id	img	DOB
▶	1	BLOB	2000-01-01
•	NULL	NULL	NULL

- Right click on the BLOB field and select “Open Values in Editor”

	id	img	DOB
▶	1	BLOB	
•	NULL	NULL	

Open Value in Editor

Set Field to NULL

Mark Field Value as a Function/Literal

Access Images in MySQL – Workbench #2

- A window will pop up, and select the “Image” tab.

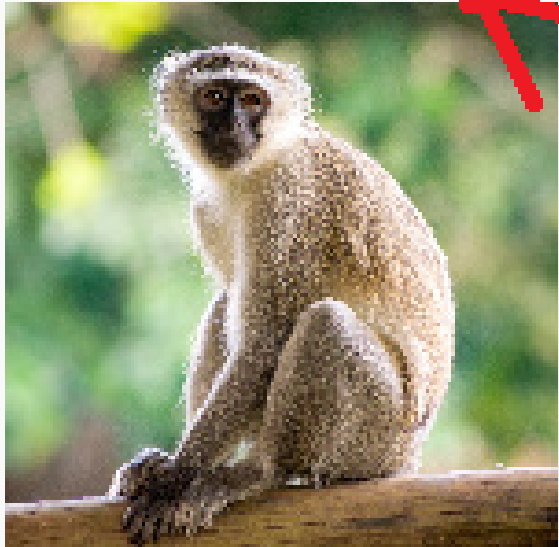
Result Grid | Filter Rows:

	id	img	DOB
▶	1	BLOB	2000-01-01
✱	NULL	NULL	NULL

Customers 4 ✕

Edit Data for img (BLOB)

Binary Text **Image**



A red arrow points to the 'Image' tab in the 'Edit Data for img (BLOB)' window.

Access Images in MySQL by PHP

- Use PHP variable to receive the image
- Display the image on the browser

```
$result = mysqli_query($con, $sql);  
$row = mysqli_fetch_array($result);
```

```
$img= $row['image'] ;  
$dob=$row['dob'];
```

```
echo "<img src='data:image/jpeg;base64,'" .  
base64_encode( $img ) . "'/>\n";
```

→ Correct

```
echo "<img src='" . $img . "'>\n";
```

→ Wrong

Tips for Project

- Please follow the project requirements to verify the functionality of your project.

Tips for project - test and verify

- You should test and verify several times.
- Make sure your program really update the values and/or delete the selected records in the database.
- If you have troubles:
 - Print the SQL query on the browser
 - Copy/paste the SQL statement and run it on Workbench
 - Do NOT retype the SQL query - must be identical
 - If the SQL query is not working on Workbench, don't expect it will work in your PHP program.
 - Possible: PHP variable is empty or 0 - primary can not be 0, wrong PK/FK, incorrect data types to the corresponding DB fields, SQL statement missing a field, etc.
- Remove the debugging message before submitting the projects.

INSERT, UPDATE, DELETE in PHP

- It is important to know and display the number of records been updated or deleted done by the SQL statement in the PHP program.
- For example: online order
 - Some items might not have enough quantity, the order can not be completed (no partial order.)
 - Customers need to know the items were really ordered
 - Need to display product name, NOT the product ID.
 - The system need to calculate and charge the real amount.
- Reference: get the number of affected rows in PHP
 - https://www.w3schools.com/php/func_mysqli_affected_rows.asp

Brainstorm exercise – Online order

- Students propose solutions
- We need to design an online order system
- Think about the real-world case – **Amazon.com**
- What tables do we need?
- What attributes should we have for each table?
- For each attribute
 - What data types?
 - What data range?
 - What constraints?

Online order – Customers table

- Here is the minimum set of attributes we need:
 - cid – customer id, long, auto increment, PK
 - Name – varchar, not null → why
 - Address – varchar, not null
 - DOB – date → do we really need it, why?
 - Sex – char → why?
 - Anything else?

Online order – Products table

- Here is the minimum set of attributes we need:
 - pid – product id, long, auto increment, PK
 - Name – varchar, not null → why?
 - Price – float or decimal(7,2) ? Range?
 - Qty – int → range?
 - When to deduct the Qty
 - Vendor information, should be name or id?
 - Anything else?

Online order – Orders table

- An “Orders” table to record all the ordering information for all the customers
- Here is the minimum set of attributes we need:
 - oid – order id, long, auto increment, PK
 - cid - customer id,
 - A FK link to Customer table → why?
 - Datetime – datetime format → why?
- Should we have payment information here?
- Should we have items information here?
 - Ordering item names, price, quantity, etc.

Online order – Order_items table

- An “Order_items” table to record all the ordering items for each order
- Here is the minimum set of attributes we need:
 - oid – order id, FK to the Orders table
 - NOT a PK → why?
 - pid – product id, FK to the Products table
 - QTY – int, is it different from Qty in Products?
 - price – decimal(7,2), should have it?
 - Will the product price be always the same?
 - What price should the system refund if the customer return one item?
 - Should we have the payment method here, why?
 - Anything else?

Check the amount for each item

- Consider the following scenario :
 - One customer adds items to their cart and takes some time before placing an order
 - Another customer may place an order during that same time frame.
 - How can we ensure that we have an enough items available for each order
 - When should we deduct the Qty from the Products table?

Operations among for these tables

- Tables
 - Customers
 - Products
 - Orders
 - Order_items
- The sequence and operation among these tables?

Exercise – Online order

- The overall flow for an online order - Amazon
 1. Customer login
 2. Search products
 3. Select and add products into the cart
 4. Repeat steps 2 and 3
 5. Check-out
 6. Enter payment method
 7. Place order
 8. Vendors ship items
 9. Customer can review order history

Last ID generated by INSERT

- When an ID is in auto increment, it is important to know the last ID generated by the INSERT statement in the PHP program.
- For example: online order
 - A customer places an order with multiple items. There should be only one order id.
- That order id will be used to
 - generated the invoice later.
 - record the customer payment information.
 - print the shipping label.
- Reference:
 - https://www.w3schools.com/php/func_mysql_insert_id.asp

Information for SQL statements – place an order

- SQL for placing an order in the Orders table
 - Customer id, date, payment methods,
 - Order id should be automatically generated by the DB
- SQL for inserting records into Order_items
 - Order id, product id, price, ordering qty,
- SQL for deducting the qty from the Products table
 - Product id
 - Ordering qty $\leq$ available qty

How to upload and insert an image in MySQL using PHP

- Create table in the database

- HTML FORM

```
<input type="file" name="image"/>
```

- PHP

```
$image = $_FILES['image']['tmp_name'];
```

```
$imgContent = addslashes(file_get_contents($image));
```

- PHP MySQL

```
mysqli_query("INSERT into images (image, created) VALUES  
( '$imgContent', curdate())");
```

- Reference

- <https://makitweb.com/upload-and-store-an-image-in-the-database-with-php/>
- <https://www.codexworld.com/store-retrieve-image-from-database-mysql-php/>

Optional

Project demo – handling multiple rows

- Must allow customer to search and
 - Display all the products that match with qty **>** 0
 - Order multiple products by one click
- Must allow employee to search and
 - Display all the products that match with qty **>=** 0
 - Update/delete multiple products by one submit
- Demo – handling multiple records

Example of sending multiple rows from front-end to back-end

- Each row might have
 - Different formats of HTML inputs
 - Some fields are updatable and some not
 - Different SQL operations for the next step
 - Different SQL data types
- Need to identify or aggregate some fields and rows

User logout

You can only update **Note** column.

ID	Code	Amount	Operation	Source	Date Time	Note	Delete
1	xy001	4000	D	ATM	2018-12-26 02:12:01	Manually inserted	☐
2	xy002	-2000	W	Online	2018-12-26 02:12:01	Manually inserted	☐
3	xy003	3000	D	Online	2018-12-26 02:12:01	Manually inserted	☐
4	xy004	-1000	W	Brarnch	2018-12-26 02:12:01	Manually inserted	☐

Total balance: 4000

Update transaction

Optional

Send multiple rows from front-end to back-end

- How to update multiple rows - HTML and PHP
 - http://imc.kean.edu/PHP_help/HTML_multiple_row_PHP.txt
- Name attribute should be array in HTML
 - Use a loop in PHP to print each record in HTML
 - A record can have several inputs.
 - Input can be hidden if we don't show them on the browser.
- Use array `$_POST['name'][$i]` to receive in PHP
 - Use a loop in PHP to receive all the records
 - The number of records is **`count( $_POST["product_id"] )`**
→ only the attribute name, NOT in array format.

Optional

Attribute in HTML array

- You can view the HTML source code of link.

http://imc.kean.edu/PHP_help/HTML_multiple_row_PHP.txt

<form>

#The inputs should be inside the loop

<input type='hidden' name='product_id[\$i]' value='3'>

<input type='text' size=4 name='cost[\$i]' value='400'>

<input type='hidden' name='user_id[\$i]' value='7'>

.....

#don't forget to increase the index **\$i++**

</form>

Optional

Receive multiple records through CGI

- In the php program, you need to use a loop and **POST** array to get all the values from HTML form. You can refer to below codes:

```
for($i=0; $i < count( $_POST["product_id"] ); $i++) {  
    $product_id[$i] = $_POST["product_id"][$i];  
    $product_name[$i] = $_POST["product_name"][$i];  
    $cost[$i] = $_POST["cost"][$i];  
  
    .....  
}
```

Know which record should be deleted or updated in the multiple rows

- In PHP, use loop to check every receiving row
- Update: Use the SQL “where” condition to check if the records has been updated
 - Compare every updatable fields value of each row receiving from HTML FORM against the values in the DB.
 - Change the database if one of the values is different.
- Delete: Use PHP to check if the delete box has been selected.
 - The multiple row index value starts from 0
 - Use the product ID from the HTML FORM to delete the selected product.

Conclusion

- How the online order works
- The tables and relationship for online order
- The sequence to interact with these tables
- Study for midterm 2